

PROMPT ENGINEERING PROJECT 1

Zero-Shot and Few-Shot Data Extraction Using Prompt Engineering

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ABSTRACT

Prompt Engineering is the process of designing effective instructions for Large Language Models (LLMs) to obtain reliable and structured outputs. This project focuses on Zero-Shot and Few-Shot Data Extraction techniques. The objective is to convert unstructured textual information into structured JSON format using carefully designed prompts.

The project demonstrates delimiter usage, schema enforcement, null fallback logic, and deterministic output generation. The developed solution ensures consistency and accuracy while extracting information such as customer names, order numbers, email addresses, and phone numbers from raw text.

Keywords: Prompt Engineering, Zero-Shot Learning, Few-Shot Learning, JSON Extraction, Large Language Models, Artificial Intelligence.

INTRODUCTION

Large Language Models have revolutionized the field of Artificial Intelligence by enabling machines to understand and generate human language. However, obtaining structured outputs from these models requires carefully designed prompts.

Prompt Engineering provides a systematic way of controlling model behavior. Instead of training a new model, prompt engineering utilizes instructions, examples, delimiters, and constraints to guide the model toward the desired output.

This project focuses on extracting structured information from unstructured customer support data using Zero-Shot and Few-Shot prompting techniques.

OBJECTIVES

The primary objectives of this project are:

1. Understand the fundamentals of Prompt Engineering.
 2. Learn Zero-Shot and Few-Shot prompting techniques.
 3. Extract structured information from unstructured text.
 4. Generate valid JSON outputs.
 5. Handle missing information using null values.
 6. Prevent prompt injection through delimiters.
 7. Create deterministic and reproducible outputs.
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THEORY

Prompt Engineering

Prompt Engineering is the process of designing and optimizing prompts to obtain accurate outputs from AI models.

A prompt consists of:

- Instructions
- Context
- Examples
- Constraints
- Output format requirements

Proper prompt engineering improves accuracy, consistency, and reliability.

Zero-Shot Prompting

Zero-Shot Prompting refers to providing only task instructions without examples.

Example:

Input: Extract name and age from the text.

Text: My name is Rahul and I am 20 years old.

Output: { "name":"Rahul", "age":20 }

Advantages:

- Simple implementation
- Lower token usage

Limitations:

- Lower accuracy
 - Less predictable formatting
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Few-Shot Prompting

Few-Shot Prompting provides multiple examples before presenting the actual task.

Example:

Input: Name: Amit

Output: { "name":"Amit" }

Input: Name: Rahul

Output: { "name":"Rahul" }

Benefits:

- Better formatting consistency
 - Higher extraction accuracy
 - Improved reliability
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JSON Data Format

JavaScript Object Notation (JSON) is a lightweight structured data format.

Example:

```
{ "customer_name":"Pradeep Kumar", "order_number":"ORD123", "email":"user@gmail.com",  
  "phone":"9876543210" }
```

JSON is widely used in APIs, databases, and enterprise systems.

METHODOLOGY

The project follows the Input-Process-Output model.

Step 1: Input

Raw customer support text is collected.

Example:

Name: Pradeep Kumar Order Number: ORD78945 Email: pradeep@gmail.com

Step 2: Processing

Prompt engineering techniques are applied:

- Delimiters
- Few-Shot Examples
- JSON Schema Constraints
- Null Handling

Step 3: Output

Structured JSON is generated.

Example:

```
{ "customer_name":"Pradeep Kumar", "order_number":"ORD78945", "email":"pradeep@gmail.com",  
  "phone":null }
```

IMPLEMENTATION

The implementation uses Python and Prompt Engineering concepts.

Tools Used:

- Python
- Streamlit
- JSON
- GitHub

Project Modules:

1. Data Input Module
2. Prompt Construction Module
3. Extraction Engine
4. JSON Formatter

RESULTS

The system successfully extracted:

- Customer Name
- Order Number
- Email Address
- Phone Number

The generated outputs were:

- Structured
 - Deterministic
 - Machine-readable
 - JSON compliant
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ADVANTAGES

1. Reduces manual data processing.
 2. Produces structured outputs.
 3. Improves consistency.
 4. Supports automation.
 5. Easy integration with databases and APIs.
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APPLICATIONS

1. Customer Support Automation
 2. Invoice Processing
 3. Resume Parsing
 4. Medical Record Extraction
 5. Banking Data Processing
 6. Enterprise Workflow Automation
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LIMITATIONS

1. Dependent on prompt quality.
 2. Complex documents may require advanced prompting.
 3. Domain-specific extraction may need additional examples.
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FUTURE SCOPE

Future improvements may include:

- Integration with GPT APIs
 - Retrieval-Augmented Generation (RAG)
 - Fine-tuned language models
 - Multi-document extraction
 - Real-time enterprise deployment
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CONCLUSION

This project successfully demonstrates the application of Prompt Engineering techniques for structured data extraction. Through Zero-Shot and Few-Shot prompting, the system converts unstructured text into valid JSON outputs. The project highlights the importance of delimiters, schema enforcement, and deterministic generation in modern AI workflows. Prompt Engineering provides a practical and efficient alternative to model training for many real-world applications.

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